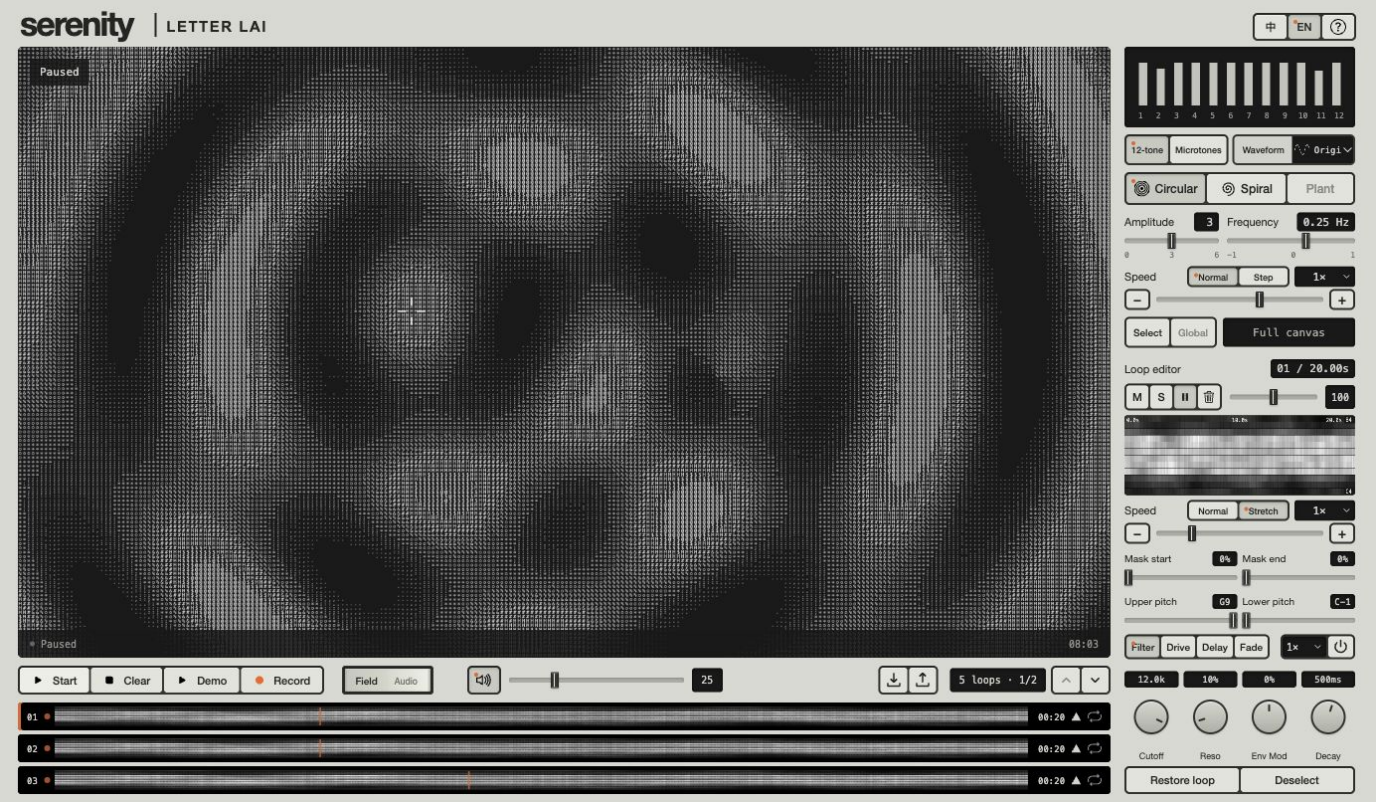
serenity

User Guide



Serenity in use

Turn an evolving field into evolving sound.

Serenity is an audiovisual instrument you play in a browser. Plant circular or spiral fields, choose where to listen, capture independent layers, then shape their rhythm, tone and echoes.

This guide assumes no prior knowledge of synthesizers or music theory. Diagrams explain the controls; screenshots show the current web version.

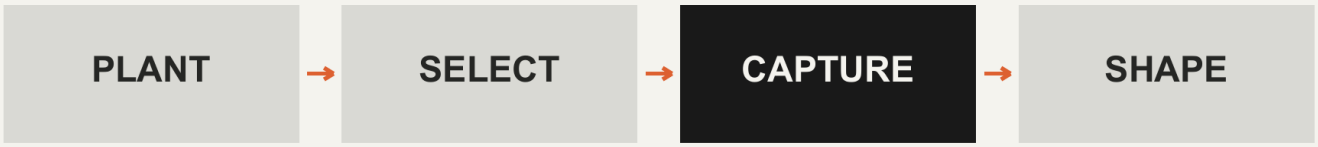
LETTER LAI · English edition

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Play online: https://laiyutao.github.io/serenity-web/

01 Your first three minutes

From an evolving field to independent sound layers.



From field to layers

1. Enable audio. Choose Begin or Demo. Start with a low master volume, then raise it gradually.

2. Choose a sound mode. Click 12-tone or Microtones in the upper panel. For a first try, choose 12-tone.

3. Plant a field. Click the screen to position the cursor, choose Circular, then press Plant. You can also double-click the screen.

4. Change one parameter. Select a source and slowly move Amplitude or Frequency. Changes spread outward rather than updating the whole screen instantly.

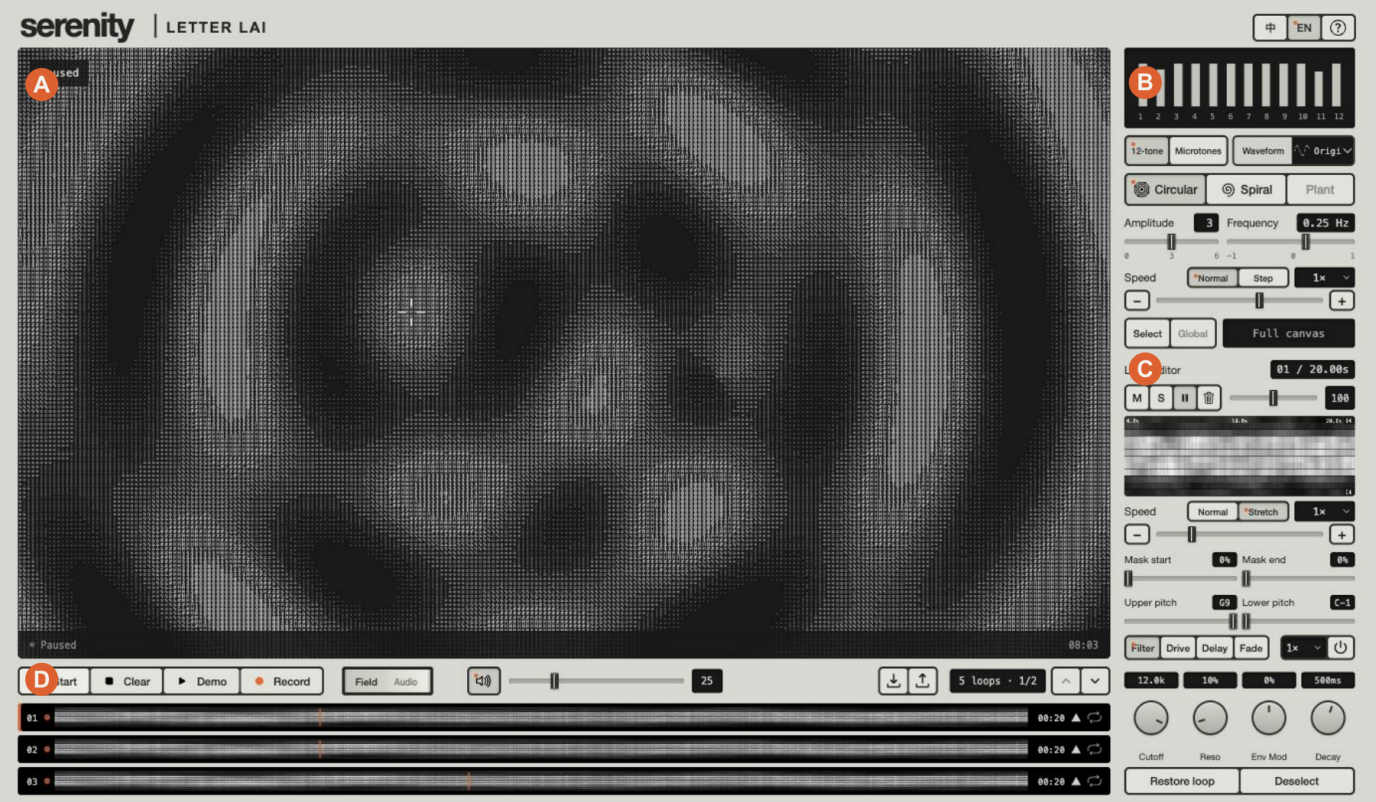
5. Listen locally. Press Select, drag a region, then press Select again to finish. Compare the selected region with Global listening.

6. Capture a layer. Select Field on the recording rocker, then press Record once (or press O). After processing, an independent loop appears below.

7. Click the upper sound mode again. After switching recording mode, or after pressing Record and generating the track, click 12-tone or Microtones in the upper panel again to resume sound from the live field. The captured track keeps playing independently.

8. Add a little space. Select the track, open Delay and add some Mix. Pause the live field to hear your captured layer on its own.

02 Find your way around



Interface map

A Main screen: fields and regions. This shows the combined field. Circular waves stay circular as the number of displayed cells adapts to the window shape.

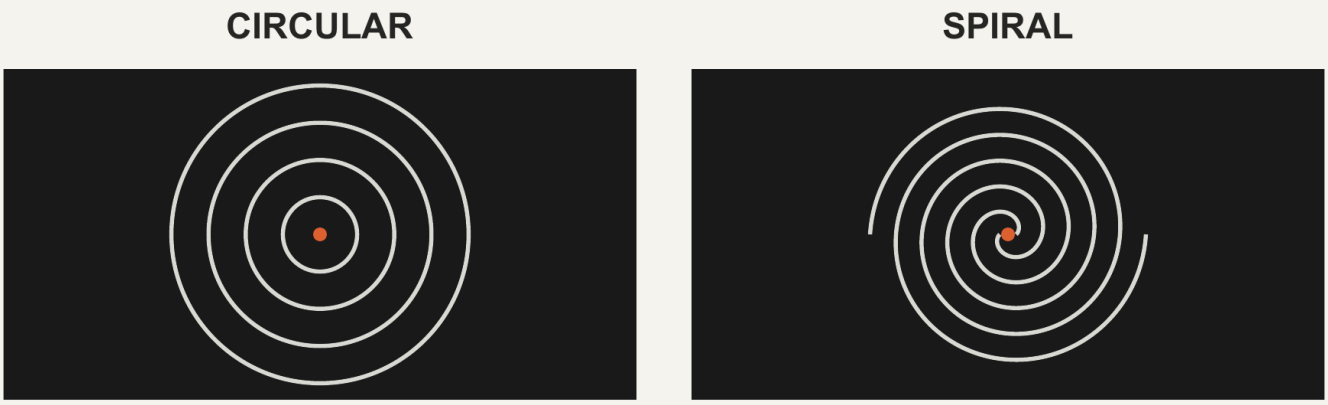
B Upper right: live sound creation. Find the pitch bars, 12-tone / Microtones, Waveform, source controls, Amplitude, Frequency, live Speed and listening regions here.

C Lower right: selected-track processing. Volume, M / S, the sound display, track Speed, masks, pitch limits and Filter / Drive / Delay / Fade affect the selected track only.

D Bottom: transport and layers. Live transport, Clear, Demo, Record, recording mode, master volume and the track list. Three tracks are shown per page; use the up/down arrows for more. The download key to the left of the track count opens audio export.

Chinese / EN changes the interface language. Hover over LETTER LAI to reveal the email address; click the name to copy it.

03 Shape a field first



Circular and spiral field diagrams

A source continuously drives the image. When fields meet, they add to or cancel one another, creating new patterns and sound. The lines above explain the shapes; the instrument itself uses grayscale characters.

Control	What it does
Circular	Expands around a point; an easy place to start.
Spiral	Angular phase differences create a rotating pattern.
Amplitude	Sets the field's strength from 0 to 6, not its track volume.
Frequency	From -1 to 1 Hz; controls the source's evolution. Zero gradually freezes reached cells.

Select an existing source to edit it. At an empty position, the settings apply to the next source you plant. A source cannot be planted twice at the same position.

Two meanings of frequency. The upper Frequency control describes field evolution, not the pitch you hear. Sound also depends on the height distribution in the listening region.

04 Choose where and how to listen



Listen only to sound produced inside the regions.

Regions and sound mapping

Select can add multiple rectangles without replacing earlier ones. Press it again to finish; Global returns to the full screen. Overlapping regions do not count the same cells twice.

A region chooses where to listen, not which sources are allowed to sound. A field outside the region can still propagate into it and change what you hear.

12-tone: height bands control the proportions of twelve fixed pitches. Their changing mixture creates chords and clusters.

Microtones: the distribution produces one continuously changing pitch, which can move between semitones.



Five waveforms

Waveform changes texture: Sine is pure, Triangle softer, Saw and Square brighter. Original follows the original voicing and uses a sine in Microtones. Bubble is round and short; Electric is bright with string-like sustain. Try Electric at Normal 1× / Step 0.5×.

05 Default stereo: hear the field's position



The same pitch reaches both ears in different proportions

With headphones, sounds naturally spread across the left and right sides. There is no extra switch or knob. The existing pitch, rhythm and timbre rules stay the same; their balance between your ears changes.

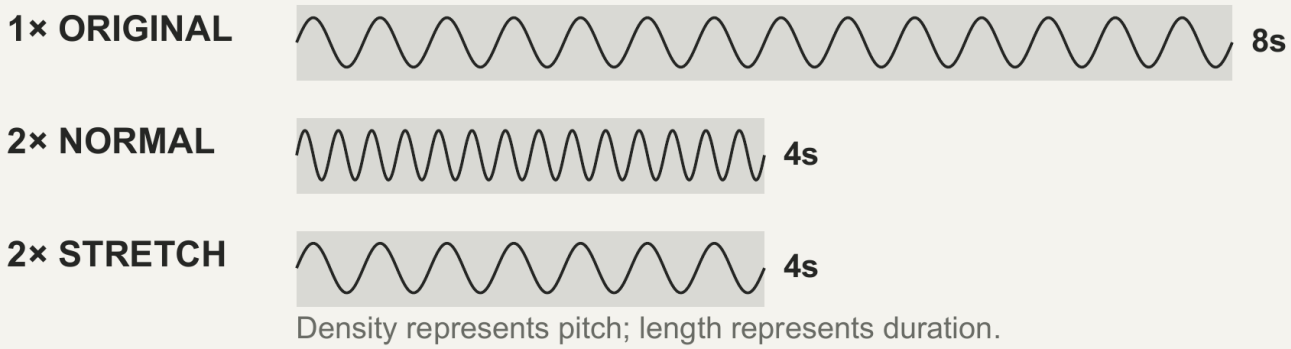
12-tone: each note has its own position. Each note follows the average horizontal position of the cells producing it. Notes toward the left are stronger in the left ear, and vice versa. A note distributed evenly across both sides stays centred. This follows the sounding cells, rather than the planted source locations.

Microtones: one moving voice. The original weighted pitch stays intact. The whole voice moves with the average position of the sounding cells, without creating two different pitches.

Global listening uses the full screen. Regions share the span from the leftmost to the rightmost valid selection; overlaps count once. Even at the edges, the other ear retains sound. Position changes are smoothed over about 40 ms. A symmetric field may remain completely centred.

Field capture, Audio recording, per-track recording, effects, Normal / Stretch, project files and WAV export preserve both channels. Old mono tracks stay centred. A single speaker makes the spatial difference less apparent.

06 Live speed and track speed



Normal versus Stretch

Upper Normal: controls live-field evolution, direction and pitch. Positive runs forward; negative reverses from the current shape without undoing sources. Saved tracks stay independent.

Upper Step: changes only how densely the field is sampled for sound, from 0.05 to 4× on a logarithmic slider. At 1× it samples every frame; at 0.25× every four frames; at 0.05× every twenty frames. Values are held between samples. It changes neither image speed nor WAV sample rate.

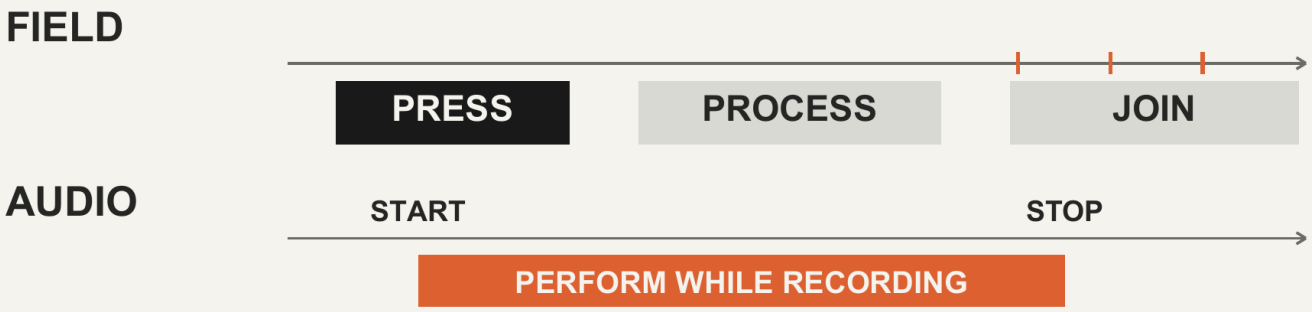
Normal and Step stay active together. Buttons only select which value to edit. Each remembers its setting; switching resets neither the sound nor the sampling clock.

Lower Normal / Stretch: affect the selected track. Each remembers an independent 0.25-4× value, and both stay active. Normal changes speed and pitch; Stretch changes speed alone. Normal 2× with Stretch 0.5× keeps the rhythm but raises pitch an octave. Switching buttons only selects the editor and keeps the playhead. The diagram's 2× examples assume the other parameter stays at 1×.

Stretching a direct recording may add texture, especially at extreme rates. Try comparing the modes around 0.75-1.5× first.

Pause is not master mute. Live transport pauses the current field and its sound, while captured tracks keep playing. Master volume and the speaker button control all output.

07 Field capture or direct recording?



Two ways to capture sound

After pressing Record, click the upper sound mode again. Once Field capture is complete, click 12-tone or Microtones in the upper panel again to resume live-field sound. The captured track keeps playing independently. In Audio mode, start recording, click the sound mode again, then continue performing.

Mode	Best for
Field	Capture the current field's sound evolution. One press computes a field cycle and compensates processing time to join the live position.
Audio	Record an actual performance. Press Record to start and again to stop; keep changing speed, waveform, regions and sources while recording.

Both modes capture only the live field, never the existing track mix. Audio mode does not record a microphone. A direct recording needs at least about half a second.

A field cycle is not one acoustic oscillation. A 0.25 Hz field repeats in 4 seconds, a 0.20 Hz field in 5 seconds. Together they may take 20 seconds to realign, while many faster audible oscillations occur inside that span.

For an easy-to-follow loop, let the field and parameter changes settle before capturing. A microtonal pitch contour can repeat without the fine audio oscillation returning to exactly the same phase at the boundary; pitch does not need to be retuned to force it.

08 Work with independent layers



Reading the sound display

Keep up to 16 tracks, with three visible per page. Click a track to show its controls on the right. Selecting another track preserves each track's settings.

The display runs through time horizontally and pitch vertically. Brighter areas indicate greater relative intensity; black indicates silence or masking. It is neither a single oscilloscope trace nor an output decibel meter.

Control	Meaning
Volume	100 is unity gain, 0 silent, 200 twice the signal gain, not necessarily twice the perceived loudness.
M / Mute	Silence this layer without deleting it.
Play / pause	Freeze this track; resume from the same position.
S / Solo	Hear only soloed recorded layers; the live field remains a separate input.
Delete	Removes the track; use M instead for temporary silence.

To audition a layer, pause the live field and use S. A muted track still keeps its display and content.

09 Duplicate, align and play one track



Copies compensate processing time and align with the original

Select first, then double-click. Click the black track row to select it. Double-click that already selected row to duplicate it. Double-clicking an unselected row only selects it; it does not immediately make a copy.

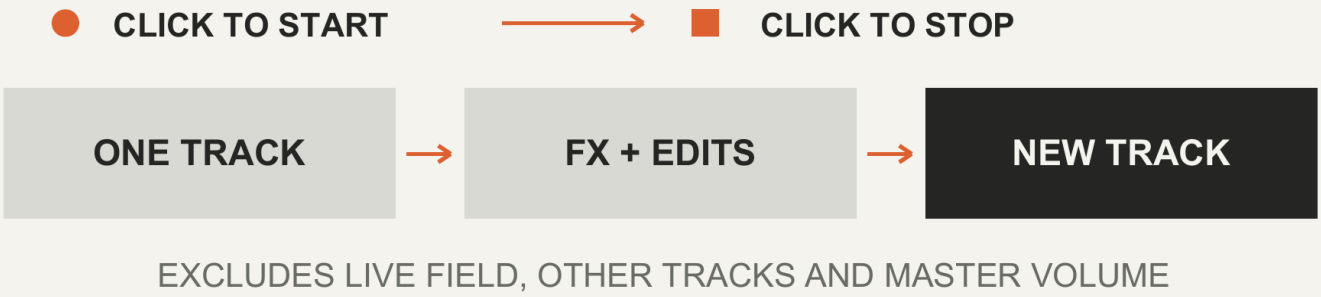
The copy appears after its source. It retains the original audio, stereo channels, masks, Normal / Stretch, volume, M / S, all four effects and any field-recall snapshot. Processing time is compensated, including oscillator phases and effect tails, so both tracks line up. You can edit them independently afterward; both tracks contribute sound.

Play / pause sits to the right of M and S. It freezes only the selected track's position and effect processing. Other tracks and the live field continue. Resume picks up where it stopped. M silences a track while its playhead keeps moving. Copying a paused track creates a paused copy.

Shift+P: pause / resume the selected track. **Shift+O:** start / stop per-track recording. Once a take is running, the next Shift+O stops that take even if you have selected another track.

P / Space still controls the live field; O still controls bottom Field / Audio recording. Performance shortcuts are ignored in inputs and dialogs. Projects retain each track's paused state. Full-cycle export includes only playing, audible tracks. The limit remains 16 tracks.

10 Record a performance of one track



Capture this track alone, including your live edits

Click the small dot beside a track number to start recording. It becomes a square of the same size. Click again to stop. Processing creates a new Audio track while the original keeps playing independently.

Only this track is captured. Its effects, masks, pitch limits, Normal / Stretch, volume and mute changes are included. Turn knobs or change speed while recording to turn your gestures into new material.

The live field, other tracks, master volume and master mute are excluded. Solo on other tracks does not affect this recording. To silence part of the take, use the source track's own M or volume.

Record at least about half a second, up to 10 minutes. At the limit, a new track is created automatically. Edits are baked into its audio; effect settings are not copied, and it has no field-recall snapshot. You can still change its speed, masks and effects afterward.

During recording and processing, the source cannot be deleted and projects cannot be saved or opened. Finish recording first. If all 16 track slots are occupied, make room before starting.

11 Make space with masks



Black sections are muted; loop length stays the same.

Ordinary and wrapping time masks

A time mask silences a section; it does not select the only section to play. Start 25% and End 50% mute the middle quarter while the rest keeps its original loop position.

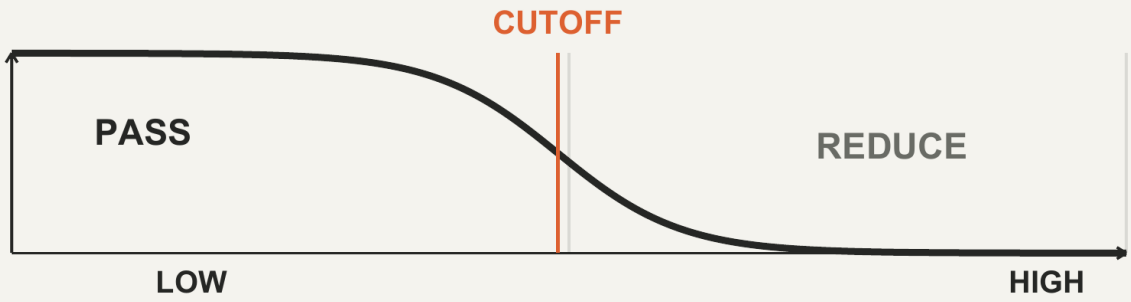
To cross the loop boundary, set Start to 75% and End to 25%. This mutes the last quarter and the first quarter. Equal start and end values mean no time mask.

You can also drag horizontally across the sound display to mark a masked interval. Its black areas follow the time mask.

Upper pitch / Lower pitch keep pitches inside the band and remove those outside it. Readouts show note names such as C4 or G9. This is pitch selection, not clipping the tops of the waveform.

Restore loop resets masks, pitch limits, track speed and effects. **Deselect** only clears the selection; it does not stop or delete the layer.

12 Filter: shape brightness



Low-pass filter diagram

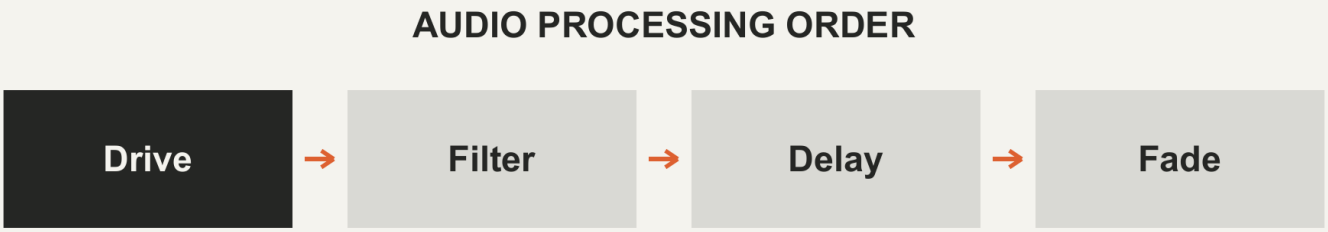
Filter is a low-pass filter: lower frequencies pass more easily while higher ones are reduced. It can soften bright harmonics or sweep automatically through the loop.

Knob	What to listen for
Cutoff	Lower values darken the tone; below the fundamental, the whole sound can become quieter.
Reso	Emphasizes the cutoff region, adding definition or a sharper, resonant character.
Env Mod	Positive sweeps brighten then darken; negative sweeps darken then brighten; zero disables the sweep.
Decay	Sets how long the sweep takes to return: short and plucky, or slow and breathing.

Start with a Saw or Square layer, set Env Mod to zero and move Cutoff alone. Learn the brightness change before adding motion. A pure sine has few harmonics, so filtering may sound mainly like a volume change.

Filter / Drive / Delay / Fade switches the control page. All four groups can work together. Moving a knob enables that group; use its power key to compare enabled and bypassed sound.

13 Drive: from saturation to fuzz



Drive processes the sound before Filter

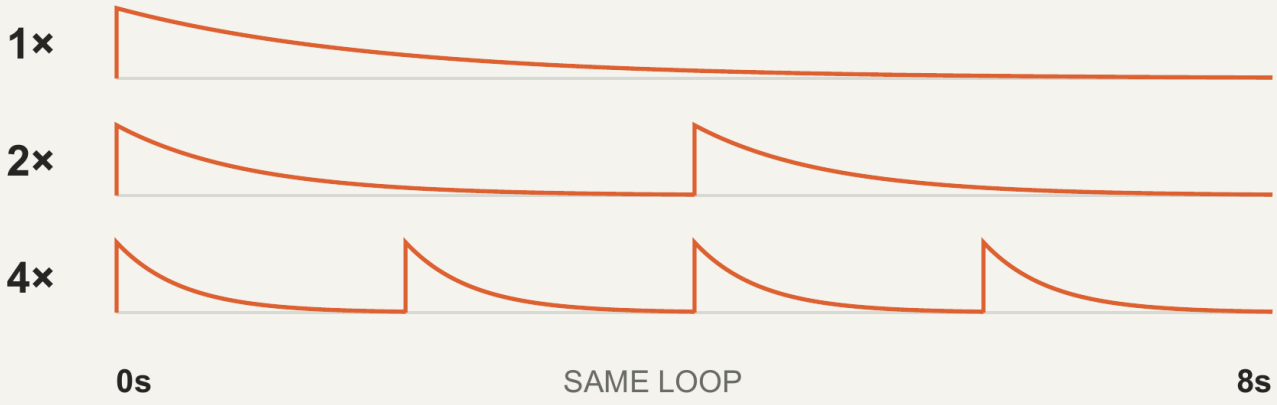
Drive adds harmonics and roughness. Small amounts can thicken or brighten a layer; higher settings add clipping and a tearing fuzz texture. It is an effect, rather than a replacement for the source waveform.

Knob	Meaning
Drive	Distortion intensity, from light saturation toward strong clipping.
Fuzz	Moves from softer shaping toward harder, rougher clipping.
Tone	Brightness of the distorted sound; lower it to tame harsh highs.
Mix	Balance of dry and distorted sound; 0% is dry.

Try an Electric, Saw or Square track with low Drive, Fuzz and Mix. Raise Drive, then adjust Tone. If the result feels too dense, lower Mix or the track's volume.

Tabs appear as Filter / Drive / Delay / Fade. Audio runs through Drive first, then Filter and Delay; Fade shapes the whole output. All four groups remember their settings and can work together. Tabs only select a control page; power bypasses that group alone. Drive starts bypassed and enables when you turn one of its knobs.

14 Repeats, not playback speed



Sweeps within one eight-second loop

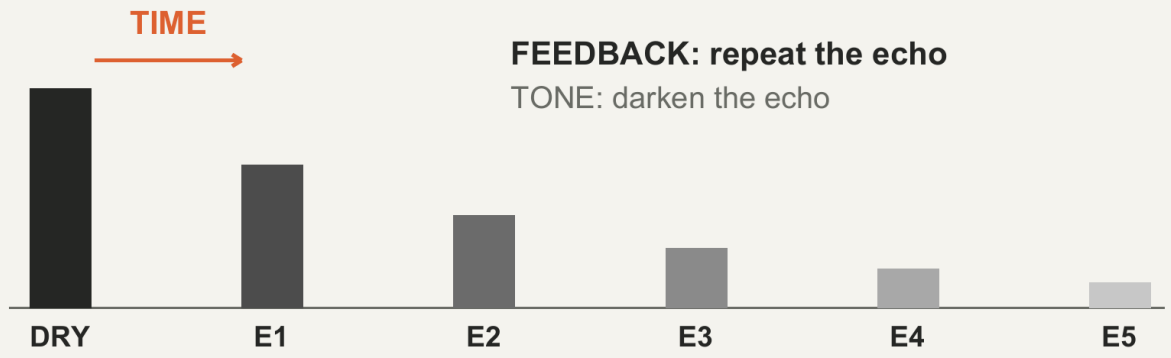
The Filter clock's 1×, 2×, 4× and 8× set **how many automatic sweeps happen during one track loop**. More repeats create denser movement without speeding up the track itself.

An 8-second loop	Sweep interval
1×	Once every 8 seconds
2×	Once every 4 seconds
4×	Once every 2 seconds
8×	Once every 1 second

Repeats sets how often a sweep begins; Decay shapes its return. At 4×, a short Decay feels pulsed and a long one more continuous. With Env Mod at zero, repeats and Decay do not create an audible automatic sweep.

Try Cutoff 250Hz, Reso 25%, Env Mod +50% and Decay 500ms. Compare 1×, 2× and 4× without changing the track speed at the same time.

15 Delay: add echoes



Echo timing and decay

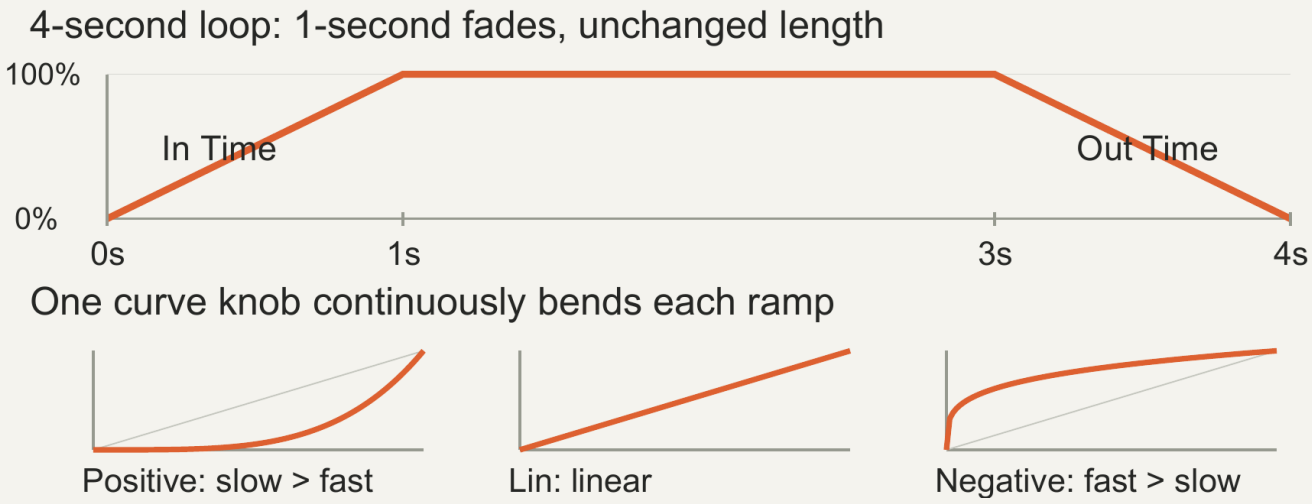
Delay plays earlier sound again after a gap. Feedback repeats the echo, and Tone changes its brightness. This is an effect tail, not another captured track.

Knob	Meaning
Time	In Free mode, sets the gap from about 10ms to 2s.
Feedback	More feedback sustains more repetitions; begin around 25%-35%.
Mix	Balance of dry and delayed sound: 0% dry only, 100% delayed only.
Tone	Darkens the echoes with a low-pass filter; it does not directly darken the dry sound.

Free uses the Time knob. **1/2, 1/4, 1/8 and 1/16** use a fraction of the whole loop. In synced mode, Time shows the calculated value and is not adjustable.

For a 4-second loop, 1/4 means 1 second; for a 20-second loop, it means 5 seconds. This is not a BPM-detected quarter note. Overlong choices are disabled; the delay limit is about 16 seconds.

16 Fade: shape volume, keep the loop



Fade length and curve are separate controls

Fade changes volume within the existing loop, without trimming, stretching or moving it. Masks stay separate. Filter, Drive, Delay and Fade can work together.

Knob	Meaning
In Time	Fade-in length; zero disables the fade-in.
In Curve	Lin = linear; + slow to fast; - fast to slow.
Out Time	Fade-out length; zero disables the fade-out.
Out Curve	Lin = linear; + fast to slow; - slow to fast.

Time follows the loop proportion. In a 4-second loop, set both ends to 1 second; at 2x speed, each becomes 0.5 seconds. The fixed 1x readout means once per cycle.

The orange line is the active volume envelope. Turning a knob enables Fade; its power key bypasses it. Fade also shapes Delay echoes. Overlapping ramps multiply, making the middle quieter.

Start at Lin, adjust times, then bend the curves. Defaults are zero-length fades and bypass.

17 Three exercises to try

REGION | FILTER | DELAY

From a dry layer to movement and space

Hear a region change

Load Demo and choose Microtones. Compare Global with a small region, then try another position. Listen for changes in pitch. Capture a Field layer and pause the live field.

Make a breathing layer

Capture a Saw or Square Field layer. Set Filter to Cutoff 250Hz, Reso 25%, Env Mod +50%, Decay 500ms and 4×. Then select the lower Stretch bank and set 0.75×, keeping Normal at 1×.

Record a gesture and echo it

Choose Audio, press Record to start, click 12-tone or Microtones again, then slowly change the upper live Speed, then stop. Add Delay: Free, 350ms, Feedback 35%, Mix 25%, Tone 3kHz. Compare Normal and Stretch track-speed modes.

Change one knob at a time. Understand one change before adding another. Use the effect power key to compare before and after without losing your settings.

18 Export your music as WAV



Live capture and full-cycle export

The download key to the left of the track count opens export. Record creates tracks inside the page; export saves your music as a portable WAV file.

Live capture. Choose Live capture and Start capture. It records the mix you hear, including the live field and saved tracks, plus changes to volume, mute, Solo, speed and effects. Click the bottom export key again to stop and download.

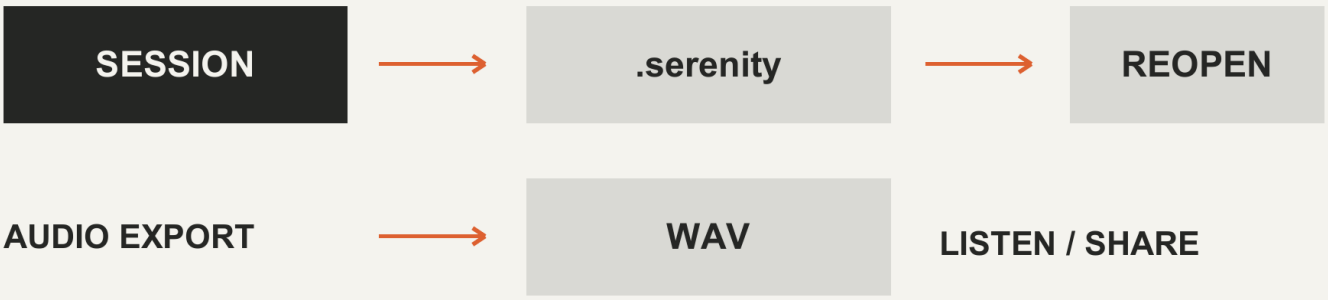
Full cycle. Choose an anchor track and Render and export. It copies the current audio state and quickly renders the common cycle from that track's next beginning, without waiting for playback. It includes audible saved tracks across all pages, excluding the live field, paused tracks, muted tracks and tracks excluded by Solo. Volume, speed, masks and Filter / Drive / Delay / Fade use their starting settings.

Truncation is explicit. Four-second and six-second loops share a 12-second cycle. If a common cycle exceeds 10 minutes, the panel and start button say truncated; only the first 10 minutes of audio are rendered. This is an audio-length limit, not a 10-minute wait. Its filename contains truncated. Live capture also ends automatically at 10 minutes.

Track editing is locked during rendering. Click the bottom download key again to cancel; cancellation downloads no partial file. Microtonal oscillations and effect tails may not join seamlessly. Gray text to the left of the download key shows export status.

O records; P controls live playback. In Field mode, O creates one track. In Audio mode, O starts recording and a second O stops it. P (or Space) pauses / resumes the live field and its sound. It does not pause saved tracks or stop export. Performance shortcuts are ignored in inputs and dialogs.

19 Save and open an editable project



WAV saves sound; a project keeps the editable session

Choose Save project in the download dialog to download a .serenity file. It includes the current field, source history, regions, sound mode, waveform, Normal / Step, plus each track's audio, stereo channels, speeds, volume, mute / Solo, masks and all four effect banks and per-track pause states.

Field tracks also retain their recall snapshots. Click a track's solid triangle to replace the live field with its captured field and listening setup. Existing tracks keep playing independently. Audio recordings and per-track recordings have no recall triangle. Later track effects and speed edits are not applied back to the field.

To open a project: click the upload key beside download and choose the .serenity file. It is read locally, with no server upload. Opening replaces the current session, so save the current project first if you want to keep it.

Projects load paused, retaining playback positions and relative track offsets. Press play to continue; effect tails restart. Old mono tracks remain centred, while new recordings preserve stereo.

WAV and projects serve different purposes. WAV is for listening and sharing; it cannot restore an editable session. Save .serenity to keep creating later. Save before refreshing or closing the page. Finish or stop recording, track processing or export before saving/opening. The file limit is 768 MB.

20 Quick reference and troubleshooting

Key	Action
W A S D	Move the planting cursor
J / K	Plant Circular / Spiral
Hold J + W / S	Adjust selected-source amplitude
Hold K + W / S	Adjust selected-source frequency
J + K	Switch 12-tone / Microtones
O	Record; press again to stop direct recording
Shift+P	Pause / resume the selected track
Shift+O	Start / stop per-track recording
P / Space	Pause / resume the live field
M	Master mute
R / X	Clear the field / Return to welcome
Esc	Exit wheel control; otherwise finish region selection

No sound? Click Begin and check master volume/mute. Make sure a field exists, or try Demo. For a silent track, inspect M / S, masks, pitch limits and Cutoff.

Why is another sound still audible in Solo? S manages captured tracks only. Pause the live field to audition a track alone.

Wheel control? Click a slider or knob to select it without changing its value. Scroll anywhere; either Shift key slows it tenfold. Click elsewhere or Esc to exit. Drag without an outline; double-click a knob to reset. Synced Delay Time is read-only.

Before reloading? Choose Save project in the download dialog, then open the .serenity file with the upload key next time. WAV saves sound only. Unsaved sessions are lost on reload or close; changing language does not clear tracks.

Contact: Hover over LETTER LAI to reveal the email, then click the name to copy it.

Play online: https://laiyutao.github.io/serenity-web/